25900235

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Value Addition Course (VAC) – MG3VACBCH200 – MICROPLASTICS AND ENVIRONMENT

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks:** 50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I), Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Multiple Choice Questions

Answer all questions
Each question carries **1 mark**

1. Microfibers are types of
   ANS: d) Microplastics [U] / [CO1]
2. Which of the following summarizes the human health risk of microplastic biomagnification?
   ANS: b) Potential endocrine disruption, inflammation, and toxicity [U] / [CO1**]**
3. The effect of microplastic in plants can cause:
   ANS: d) Impaired root growth and function [U] / [CO2]
4. Which uncertainty is biological in nature?
   ANS: a) Unpredictable interactions with body [K] / [CO3]
5. What happens when polar sea ice melts in relation to microplastics?
   ANS:b) Microplastics are released into the oceans [K] / [CO4]

[1×5 = 5]

Part B – Fill in the Blanks

Answer all questions
Each question carries **2 marks**

6. The growing concern about microplastics is mainly due to their potential impacts on environmental and _____.
   ANS: Human Health [U] / [CO1]
7. Microbeads are plastic particles smaller than _____ millimeters.
   ANS: 5mm [K] / [CO1]

8. Microplastics negatively impact the _____ industry by reducing seafood safety and availability.
   ANS: Sea foodindustry [K] / [CO2]
9. Microplastics can reduce _____ quality in males, affecting fertility.
   ANS: Sperm [K] / [CO3]
10. In adsorption method, _____ and _____ are used as adsorbents for removing microplastics.
    ANS: Activated carbon, Graphene, biochar [U] / [CO4]

[2×5 = 10]

Part C
Short Answer Questions

Answer any 5 questions

Each question carries **3 marks**

11. Explain how rivers act as pathways in transporting microplastics to oceans.
    ANS: Write any three points (each point carries one mark) [U] / [CO1]
12. Define bioaccumulation. Name one heavy metal that undergoes bioaccumulation in aquatic organisms.
    ANS: Definition 2.0 marks
    Name any one heavy metal ex.Mercury 1.0 mark [K] / [CO1]
13. Discuss the significance of classifying microplastics into different categories.
    ANS: Classification 1.5 marks
    Significance 1.5 marks [U] / [CO1]
14. List any three key effects of microplastics on freshwater ecosystems.
    ANS:Write any three points (each point carries one mark) [U] / [CO2]
15. Recall the sources of microplastics in drinking water.
    ANS:Write any three points (each point carries one mark) [K] / [CO3]
16. Mention the areas where safety limits for microplastics are set by regulatory agencies.
    ANS:Write any three points (each point carries one mark) **[**K] / [CO4]
17. Explain how microplastics cause habitat disruption and alter food webs in marine ecosystems.
    ANS:Explain habitat disruption by microplastics – 1.5 marks
    Explain food web alteration in marine ecosystems by microplastics -1.5marks
    [U] / [CO3]

[3×5 = 15]

Part D
Short Essay Questions

Answer any 4 questions

Each question carries 5 marks

18. Demonstrate the possible mechanisms to minimise the human exposure to microplastics.
    ANS: ANS: Write any FIVE points (each point carries one mark) [U] / [CO1]
19. Assess the effectiveness of chemical versus biological methods for microplastic degradation, highlighting their advantages and limitations.
    ANS: Chemical method: Degradation, advantages and limitation (2.5marks)
    Biological method: Degradation, advantages and limitation (2.5marks)
    [E] / [CO1]

20. Evaluate the different types of physiological, behavioral, and ecological responses exhibited by wildlife to microplastic exposure.
    ANS: Explain
    Physiological responses (2 marks)
    Behavioral responses (1.5 marks)
    Ecological responses (1.5marks) [E] / [CO2]
21. Give the impact of different types of pollutants on microplastics distribution in food.
    ANS: Explain the impact of any 5 number of pollutants on microplastics distribution in food ((each point carries one mark) [U] / [CO3]
22. Discuss how microplastics can alter soil structure and fertility.
    ANS: Any five points (each point carries one mark) [U] / [CO1]
23. Evaluate the impact of microplastics on the cryosphere, microbial and biogeochemical processes, focusing on zooplankton and carbon cycling in marine ecosystems.
    ANS: Explain the impact of microplastics on
    a. **Cryosphere (1.5 marks)**
    b. **Microbial (1.5 marks)**
    c. **Biogeochemical (2 marks)** [E] / [CO4]